

Major Project Report
On
Generation of Deformation Mechanism Maps of 2219
Aluminium Alloy and Aluminium Alloy micro-alloyed
with 0.1 wt% Silver

Submitted in partial fulfilment of the requirements for the award of the
degree of

Bachelor of Technology
in
Mechatronics Engineering
By

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June 2026



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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR
(An Institute of National Importance under Ministry of Education, Govt. of India)

DECLARATION

We hereby declare that the work reported in this project titled "***Generation of Deformation Mechanism Maps of 2219 Aluminium Alloy and Aluminium Alloy micro-alloyed with 0.1 wt% Silver*** " is the original and has been carried out by us independently in the Department of Mechatronics Engineering, IIIT Bhagalpur under the supervision of Dr. Purnendu Kumar Mandal, Assistant Professor, Department of Mechatronics and Automation Engineering, IIIT Bhagalpur. We also declare that this work has not been the basis for the award of any other degree, Diploma, or similar title of any university or institution and no any AI tool been used while writing this report.

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CERTIFICATE

This is to certify that the project entitled " *Generation of Deformation Mechanism Maps of 2219 Aluminium Alloy and Aluminium Alloy micro-alloyed with 0.1 wt% Silver* " is carried out by **Nilansh Singh (2201201ME)** and **Vikas Kumar (2201223ME)** B.Tech undergraduate students of IIIT Bhagalpur, under my supervision and guidance. This project has been submitted in partial fulfilment of the requirements for the award of "*Bachelor of Technology*" degree in *Mechatronics Engineering* at *Indian Institute of Information Technology, Bhagalpur*.

No part of this project has been submitted for the award of any other degree or diploma to the best of my knowledge.

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ACKNOWLEDGEMENT

With great pleasure we express our cordial thanks and indebtedness to our project supervisor, *Dr. Purnendu Kumar Mandal*, Assistant Professor, Department of Mechatronics Engineering. His vast knowledge, expert supervision and enthusiasm continuously challenged and motivated us to achieve our goal. We will be eternally grateful to him for allowing us the opportunity to work on this project.

We express our sincere gratitude to Dr. Abhinav Gautam, Assistant Professor and Head of Department, ME for his valuable help, guidance and all the relevant facilities that have made the work completed in time.

During the course of this project report preparation, we have received lot of support, encouragement, advice and assistance from many people and to this end we are deeply grateful to them all.

We have great pleasure in expressing our sincere gratitude and thanks to the *Prof. Madhusudan Singh*, Director, Indian Institute of Information technology Bhagalpur for his constant encouragement and support in completion of this innovative project work.

We take this opportunity to thank all faculty members of department of Mechatronics and Automation, IIIT Bhagalpur for their persistence in academic excellence throughout the years.

The present work certainly would not have been possible without the help of our friends, and also the blessings of our parents.

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ABSTRACT

The high-temperature deformation behaviour of metallic alloys is governed by complex interactions between strain, strain rate and deformation temperature. Understanding this behaviour is essential for optimising thermo-mechanical processing operations such as forging, rolling and extrusion. In the present work, Artificial Neural Network (ANN)-based models were developed to predict the flow stress behaviour of 2219 aluminium alloy (Alloy A) and 2219 aluminium alloy micro-alloyed with 0.1 wt.% silver (Alloy B) under varied hot deformation conditions. The experimental dataset used for model development was drawn from isothermal compression tests carried out over a strain rate range of 10^{-3} to 10^1 s^{-1} and a temperature range of 300–500°C. The Defect region is observed at low temperature and high strain rate region.

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Chapter 1: Introduction

1.1 Overview

Aluminium alloys from the 2xxx series have earned a strong reputation in aerospace, defence and high-performance structural work, mainly because they offer a useful mix of light weight, impressive strength-to-weight ratio and decent ductility even when things get hot. The 2219 aluminium alloy stands out among them — it uses copper as its primary strengthening agent and is widely considered one of the more flexible heat-treatable options in this family. It welds reliably, holds up well at extremely low temperatures and still delivers adequate strength under elevated service conditions, which is why engineers keep reaching for it when designing cryogenic fuel tanks, spacecraft structural panels and aircraft fuselage sections. [1, 2].

Before a wrought aluminium alloy can be used in any structural application, it needs to go through one or more thermo-mechanical processing steps such as hot forging, extrusion or rolling. These operations do more than just shape the material — they also clean up the microstructure by breaking down segregations and internal defects that were left behind during casting. Getting these steps right depends heavily on choosing the correct combination of deformation temperature and strain rate, because the alloy behaves quite differently depending on the conditions it is processed under. At certain temperature and strain rate combinations, useful things happen inside the material — dynamic recovery and dynamic recrystallisation kick in, softening the alloy and encouraging the formation of fine-grained microstructures. But at other combinations, things can go wrong. Adiabatic shear bands may form, wedge cracks can develop, or ductile fracture can set in, all of which result in defective products that fail to meet quality standards. [3, 4].

The relationship between flow stress, strain, strain rate and temperature during hot deformation is nonlinear and complex. Traditional constitutive equations, most notably the hyperbolic-sine Arrhenius type, capture this relationship with reasonable accuracy for many alloys. P.K. Mandal et al. [5] applied this approach to study the hot deformation behaviour of 2219 Al alloy and a variant micro-alloyed with 0.1 wt.% silver, establishing constitutive models and computing activation energies for both systems. Their work showed that the addition of silver lowered the flow stress and reduced the activation energy for deformation, while the predicted peak flow

stresses agreed with experimental values within a deviation of about 10% for 92% of the data points.

However, constitutive equations require a substantial number of material constants, and the fitting procedure can be cumbersome when the dataset is large or when the relationship is highly nonlinear across a wide processing window. Artificial Neural Network (ANN) models offer an alternative approach: instead of assuming a functional form, the network learns the input–output mapping directly from data. ANN models have been applied successfully to predict the flow stress of various aluminium alloys, steels and magnesium alloys, and have demonstrated prediction accuracies comparable to or exceeding those of conventional constitutive equations [6–13].

Once a reliable ANN model is in place, it can generate flow stress values at any desired combination of deformation conditions, including conditions not covered by the original experimental programme. These predicted datasets are particularly useful for constructing Deformation Mechanism Maps (DMM) and processing maps within the Dynamic Materials Model (DMM) framework, which require flow stress data over a dense grid of temperatures and strain rates [14–16]. Processing maps divide the temperature–strain-rate space into domains associated with specific deformation mechanisms, thereby guiding the selection of optimal hot working parameters.

The present work addresses the complete workflow developing and validating ANN-based flow stress models for Alloy A and Alloy B, using the optimised models to generate extended datasets and construct deformation mechanism maps for both alloy systems.

1.2 Literature Survey

The way aluminium alloys behave under hot deformation has been a subject of serious research interest for several decades now, largely driven by the ongoing need to fine-tune thermo-mechanical processing across industries ranging from aerospace to automotive manufacturing. The review that follows traces the key developments in this space, covering how researchers have experimentally characterised these alloys, how constitutive models have been built and refined to describe their behaviour, and how machine learning has more recently been brought into the picture to predict flow stress with greater accuracy and efficiency.

Early investigations into the high-temperature flow behaviour of 2xxx-series aluminium alloys established the general framework still used today. Jonas et al. [3] showed that the Arrhenius-type hyperbolic sine equation, originally developed for creep, could describe the strain-rate and temperature dependence of flow stress in hot-worked metals with sufficient accuracy. This equation, characterised by the activation energy Q and the Zener–Hollomon parameter Z , became the standard constitutive model for hot deformation analysis.

For 2219 aluminium alloy specifically, Kaibyshev et al. [2] carried out a detailed study of deformation behaviour over a range of temperatures and strain rates, reporting that the stress exponent takes values of 7 and 5 in the warm and hot deformation ranges respectively, consistent with a dislocation climb mechanism. Zhang et al. [17] extended this work by examining the effect of initial microstructure on the hot compression behaviour of 2219 Al alloy, finding that the homogenised condition produced lower flow stresses and more uniform deformation compared to the as-cast condition.

The effect of microalloying on deformation behaviour has been examined for several aluminium alloy systems. Li et al. [18] investigated the hot deformation behaviour of a 2519 aluminium alloy containing silver and reported that Ag addition led to changes in precipitate morphology and flow stress, findings that are conceptually relevant to the alloy system examined in the present work. P.K. Mandal et al. [5] carried out the most directly relevant study: they performed isothermal hot compression tests on Alloy A and Alloy B at temperatures of 300–500°C and strain rates of 10^{-3} to 10 s^{-1} , developed constitutive equations for both alloys and demonstrated that silver addition decreased the activation energy for deformation by approximately 5%.

ANN-based prediction of hot deformation behaviour has grown significantly since the pioneering work of Robi and Dixit [55 in Ishaan's ref], who applied neural networks to generate processing maps for a hot working application. Sabokpa et al. [10] and Mandal et al. [11] demonstrated that feed-forward back-propagation ANN models with the Levenberg–Marquardt training function could predict flow stress of an AZ81 magnesium alloy and AISI 304L stainless steel respectively, achieving accuracies comparable to conventional constitutive equations. Haghdadi et al. [12] applied a similar approach to an A356 aluminium alloy, confirming that ANN models generalise well to unseen deformation conditions when properly trained and validated.

For the 2xxx-series aluminium alloys, Ji et al. [9] carried out a comparative study between Arrhenius-type constitutive equations and ANN models for Aermet100 steel and reported that ANN models produced lower prediction errors across most deformation conditions. Quan et al. [8] similarly found ANN models superior for predicting the highly nonlinear flow behaviour of a 3003 aluminium alloy at high temperatures and strain rates. Ashtiani and Shahsavari [6] compared phenomenological and ANN models for an AlCuMgPb alloy and concluded that ANN models were more reliable when the deformation dataset covered a wide range of conditions.

Building Deformation Mechanism Maps and processing maps, which are grounded in the Dynamic Materials Model originally developed by Prasad et al., has become a widely accepted and standard approach for optimising hot working schedules in practice. Ravichandran et al. [15] applied processing maps to pure aluminium and identified domains associated with dynamic recrystallisation. Gegel et al. [16] extended the DMM framework to include stability criteria, allowing unstable deformation domains to be identified and avoided. Raj et al. [19] provided theoretical support for the efficiency of power dissipation concept embedded in the DMM. In recent applications, the dense flow stress datasets needed for constructing high-resolution processing maps have been obtained from ANN models rather than from exhaustive experimental programmes, as demonstrated by Sun et al. [20] for a Ti600 alloy and by Dong et al. [21] for Al–Mg–Si alloys.

1.3 Research Gap

Although P.K. Mandal et al. [5] have already established constitutive equations for the Alloy A and Alloy B systems and demonstrated acceptable prediction accuracy, two important aspects remain unaddressed. First, ANN-based modelling has not been applied to these specific alloy compositions, and it is not known whether ANN models would offer improved prediction accuracy or broader applicability compared to the Arrhenius-type constitutive equations already available. Second, no Deformation Mechanism Maps have been generated for either alloy system, leaving the safe and unsafe deformation domains uncharted.

The construction of processing maps requires flow stress values on a dense grid of temperature–strain-rate conditions, which in practice is difficult to obtain through experimental testing alone. ANN models, once validated, can generate these dense datasets efficiently, making them a natural tool for feeding into DMM calculations. The absence of this analysis in the existing literature

represents a meaningful gap, given the industrial relevance of 2219-series aluminium alloys for aerospace and defence applications.

Additionally, the effect of silver microalloying on the hot workability of 2219 Al alloy, as reflected in the processing maps, has not been examined. The constitutive modelling work of P.K. Mandal et al. [5] showed that silver lowers the flow stress and reduces the activation energy, which would be expected to influence the efficiency of power dissipation and the extent of the safe working domain. Quantifying this effect through processing maps would provide useful guidance for the industrial processing of silver-micro-alloyed 2219 alloys.

1.4 Objectives

Based on the research gap identified above, the present work sets out to accomplish the following goals:

1. To design and validate ANN-based models that can reliably predict how the flow stress of Alloy A and Alloy B behaves when both alloys are subjected to varying hot deformation conditions.
2. To get the best out of the ANN architecture, the hidden neuron counts and transfer function combinations were varied and tested, while the performance of each resulting model was judged against standard statistical error measures.
3. To use the optimised ANN models to generate richer and more comprehensive flow stress datasets that cover a much denser grid of temperature and strain rate conditions than what was originally available.
4. To build Deformation Mechanism Maps and contour processing maps for both alloy systems by drawing on the generated datasets and the Dynamic Materials Model framework.
5. To pinpoint and compare the safe and unsafe hot deformation domains for Alloy A and Alloy B, and to get a clearer picture of how the addition of silver influences the overall hot workability of these alloys.

Chapter 2: Methodology

The methodology adopted in the present work can be divided into four main stages: (i) dataset preparation and preprocessing, (ii) ANN architecture design and training, (iii) testing and validation of the trained networks, and (iv) generation of extended datasets and construction of Deformation Mechanism Maps. Each stage is described in the following subsections.

2.1 Material and Experimental Details

The experimental data used in the present work were taken from the investigation reported by P.K. Mandal et al. [5], in which two aluminium alloy compositions were prepared by casting: 2219 Al alloy (designated here as Alloy A) and 2219 Al alloy containing 0.1 wt.% silver (designated as Alloy B). Both alloys were homogenise-annealed at 510°C for 10 hours followed by furnace cooling to minimise segregation and coring effects arising during the casting process.

Cylindrical compression specimens of 10 mm diameter and 15 mm height were machined from the homogenised rods. Two concentric grooves of 0.2 mm depth were machined on the end faces of each specimen to retain the MoS₂ lubricant used during testing. Uniaxial isothermal compression tests were conducted on a 100 kN servo-hydraulic testing machine (Instron Model 8801) at constant true strain rates and different temperatures mentioned in table 2.1, with the specimen held at the test temperature for 15 minutes prior to deformation to ensure thermal homogeneity. Temperature was maintained within $\pm 3^\circ\text{C}$ using a Eurotherm controller. Tests were conducted up to a maximum true strain of 0.6.

Table 2.1: Chemical composition (wt.%) of Alloy A and Alloy B

Constant Strain Rate (s^{-1})	Test Temperature ($^\circ\text{C}$)
10^{-3} , 10^{-2} , 10^{-1} , 10^0 , and 10^1	300, 350, 400, 450 and 500

2.2 Flow Stress Behaviour

The true stress–true strain curves obtained from the compression tests exhibit the general characteristics typical of hot-deformed aluminium alloys. At lower temperatures, the flow curves for both alloys show a continuous increase in stress up to a peak value, reflecting the dominance

of dislocation generation and accumulation over dynamic softening mechanisms. As the temperature rises, the driving force for dynamic recovery increases and, at the highest temperatures tested, the flow stress reaches a peak and then decreases gradually with further strain, indicating a transition towards flow softening attributed to dynamic recrystallisation and subgrain growth [5, 22].

For a given temperature, the peak flow stress increases monotonically with strain rate in both alloys, consistent with the general understanding that higher strain rates leave less time for thermally activated recovery processes. Alloy B consistently shows lower flow stresses than Alloy A under the same deformation conditions. This reduction is attributed to the substitutional role of silver atoms in the aluminium matrix, which reduces dislocation density, as confirmed by transmission electron microscopy in earlier work on Al–Cu–Mg–Ag alloys [23].

The peak flow stress versus temperature plots for both alloys show a roughly linear decrease with increasing temperature at each strain rate level, which is consistent with the observations reported by P.K. Mandal et al. [5]. The maximum peak flow stresses in the present dataset occur at the lowest test temperature (300°C) and the highest strain rate (10 s⁻¹), reaching approximately 260 MPa for Alloy A and a somewhat lower value for Alloy B. At 500°C and 10⁻³ s⁻¹, the peak flow stresses fall below 100 MPa for both alloys.

The normalization relationship used in the present work is given below as eq.no (1) :

$$x_n = 0.1 + 0.8 \times \left[\frac{x - x_{min}}{x_{max} - x_{min}} \right] \dots\dots\dots (1)$$

where:

- x_n = normalized value
- x = original experimental value
- x_{min} = minimum value of parameter
- x_{max} = maximum value of parameter

The normalized dataset was then divided into separate groups for training, testing and validation before ANN modelling.

2.3 Constitutive Modelling Background

The constitutive modelling work of P.K. Mandal et al. [5] provides an important reference for the present investigation. The hot deformation behaviour of the two alloys was described using the hyperbolic-sine Arrhenius equation as eq.no (2):

$$\dot{\epsilon} = A [\sinh(\alpha\sigma)]^n \exp(-Q / RT) \quad \dots\dots\dots (2)$$

where $\dot{\epsilon}$ is the strain rate, σ is the flow stress, Q is the activation energy for deformation, R is the universal gas constant, T is the absolute temperature, and A , α and n are material constants. For Alloy A, the activation energy was determined to be 321 kJ/mol, while for Alloy B it was 304 kJ/mol, indicating that silver addition reduces the energy barrier for deformation by approximately 5.3% [5].

The Zener–Hollomon parameter as per eq.no (3) was used to correlate flow stress with deformation conditions. The results were encouraging, with 92% of the peak flow stress values for both alloys landing within a deviation error of $\pm 10\%$ when the developed constitutive equations were used for prediction, suggesting the approach works reasonably well. However, things get complicated fairly quickly with constitutive equations, since they demand multiple constants and lengthy fitting procedures that become increasingly difficult to handle as the dataset grows to cover a wider range of conditions — and that frustration is honestly what pushed the present work towards giving ANN models a proper try.

$$Z = \dot{\epsilon} \exp(Q/RT) \quad \dots\dots\dots (3)$$

For generation of Deformation Mechanism Maps, the Dynamic Materials Model (DMM) framework developed by Prasad et al. [14] is adopted. In this framework, the workpiece during plastic deformation is treated as a power dissipater. The total instantaneous power per unit volume, P , is partitioned as follows shown as eq. no (4):

$$P = G + J = \sigma \dot{\epsilon} \quad \dots\dots\dots (4)$$

where G represents the power dissipated through heat generation and J represents the power dissipated through microstructural changes such as dynamic recrystallisation and phase transformations. The dissipater co-content J is related to G through the strain rate sensitivity parameter m of the flow stress as shown in eq.no.(5) :

$$m = \partial \ln \sigma / \partial \ln \dot{\epsilon} \quad \dots\dots\dots (5)$$

The efficiency of power dissipation η , which characterises the relative rate of microstructural change, is then given by eq. no (6):

$$\eta = 2m / (m + 1) \dots\dots\dots (6)$$

Regions of high η in the temperature–log strain rate space are generally associated with dynamic recrystallisation or superplastic flow, which are considered safe deformation mechanisms. The instability parameter $\xi(\dot{\epsilon})$, introduced by Narayana Murty et al. [33, from Ishaan], identifies regions of plastic flow instability as shown in eq.no (7) :

$$\xi(\dot{\epsilon}) = \partial \ln[m/(m+1)] / \partial \ln \dot{\epsilon} + m \dots\dots\dots (7)$$

When $\xi(\dot{\epsilon}) \leq 0$, the flow is treated as unstable, and any deformation happening in that regime should be avoided if defect formation is to be prevented. The processing map, which is put together by overlaying the efficiency contour map on top of the instability map, then serves as a thorough and practical guide for anyone trying to select the right hot working conditions [14, 16, 24].

2.4 Dataset Preparation and Preprocessing

The raw experimental dataset consisted of 150 input–output data pairs for each alloy, corresponding to six strain values (0.1, 0.2, 0.3, 0.4, 0.5 and 0.6), five strain rates (10^{-3} , 10^{-2} , 10^{-1} , 10^0 and 10 s^{-1}) and five temperatures (300, 350, 400, 450 and 500°C). The three deformation parameters (strain ϵ , strain rate $\dot{\epsilon}$ and temperature T) served as ANN inputs, and the corresponding flow stress σ was the output.

Because the strain rate values span several orders of magnitude, direct use of the raw strain rate values would create a severe numerical imbalance among the input variables and impede ANN convergence. To address this, the strain rate values were converted to their common logarithm ($\log_{10} \dot{\epsilon}$) before being passed to the network. This logarithmic transformation compresses the large numerical range and makes the relationship between input and output more uniform across the full processing window [10, 12].

All input variables were then scaled down to sit within the range [0.1, 0.9] using a straightforward linear min–max transformation. This normalization process ensures that all input variables remain within a similar numerical range, preventing any single parameter from dominating the learning process and helping the ANN achieve more stable and accurate training. [25, 26].

The output flow stress values were similarly normalised before training and de-normalised before reporting results.

The 150 data pairs for each alloy were divided into three subsets: 75 pairs (50%) for training, 45 pairs (30%) for testing and 30 pairs (20%) for validation , discussed detailed in 3.1.

2.5 ANN Architecture and Training

A feed-forward multi-layer perceptron (MLP) architecture was adopted for the ANN model. The input layer contained three neurons corresponding to the three deformation parameters (ϵ , $\log \dot{\epsilon}$, T), and the output layer contained a single neuron representing the flow stress σ as shown in table (2.2). A single hidden layer was used, as this is sufficient to approximate any continuous function to arbitrary accuracy according to the universal approximation theorem [26, 27].

Two transfer function combinations were evaluated for the hidden layer: tansig–purelin and logsig–purelin. The purelin (linear) function was always used in the output layer to allow unrestricted flow stress prediction. The number of hidden neurons was varied from 4 to 9 in a systematic trial-and-error study, giving a total of 36 ANN architectures for each alloy (two transfer function options $\times$ six neuron counts $\times$ three random weight initialisations per configuration).

Training was carried out using the Levenberg–Marquardt backpropagation algorithm (trainlm function in MATLAB Neural Network Toolbox) with an error goal of 0.000018. The Levenberg–Marquardt method is a second-order optimisation algorithm that combines the speed advantages of the Gauss–Newton method with the stability of steepest-descent gradient methods, making it particularly effective for training networks on small-to-medium-sized datasets [19, 25, 28, 29].

Different ANN architectures were developed separately for Alloy A and Alloy B using different hidden neuron combinations and transfer functions. The performance of each ANN model was compared using RMS validation value, training error and testing error in order to identify the optimized ANN structure. It was observed that the prediction capability of the network strongly depended on hidden neuron selection and transfer function combination. The optimized ANN models showed better agreement between experimental and predicted flow stress values with comparatively stable validation behaviour.

Table 2.2: ANN training parameters used in the present work

Network Parameter	Value
Architecture	Feed-forward Back Propagation MLP
Training Algorithm	Back-propagation (Levenberg–Marquardt)
Training Function	trainlm
Input Layer Neurons	3 (ϵ , $\dot{\epsilon}$, T)
Hidden Layer Neurons	4 to 9 (varied by trial)
Output Layer Neurons	1 (σ)
Transfer Functions Tested	tansig/purelin and logsig/purelin
Error Goal	0.000018

2.6 Testing and Validation

After each training run, the network was immediately evaluated on the separate testing dataset to obtain the testing error. Only models that showed consistent behaviour during both training and testing were advanced for final validation. The performance criterion for selecting the optimised architecture was based on three metrics: (i) root mean square deviation (RMSD) computed over the validation dataset, (ii) maximum percentage error in the validation set, and (iii) percentage of validation data points falling outside a $\pm 5\%$ deviation band.

The RMS functional error is defined as eq.no (8) :

$$\text{RMSD} = \sqrt{[\sum_{i=1}^n (\sigma_i - \sigma_i^*)^2 / n]} \dots \dots \dots (8)$$

where σ is the experimental flow stress value, σ^* is the ANN-predicted value and n is the number of data points in the evaluation subset [10, 13]. An ANN architecture was considered acceptable when the maximum validation error was below 5% and the RMSD was minimised simultaneously. Models exhibiting lower training RMSD but higher testing or validation errors were identified as overfitted and were discarded.

Chapter 3: Results and Discussion

This chapter presents the results obtained from ANN modelling and Deformation Mechanism Map analysis for Alloy A and Alloy B. The results cover ANN architecture selection, validation performance, error analysis for experimental flow stress value as given in fig no (2.1), processing map construction and a comparative discussion of the two alloy systems.

3.1 Selection of Dataset

The experimental deformation dataset used in the present work was obtained from hot compression testing carried out under different hot working conditions for both Alloy A and Alloy B. Each alloy consisted of 150 experimental data points generated using different combinations of strain, strain rate and deformation temperature conditions. Six different strain levels, five deformation temperatures and five strain-rate conditions were considered during the investigation. The deformation temperature range varied from 573 K to 773 K, while the strain-rate conditions ranged from 10^{-3} to 10^1 s^{-1} . Such a wide range of deformation conditions helped in developing a more reliable ANN model for predicting flow stress behaviour under different thermo-mechanical processing conditions.

The 150 data pairs for each alloy were divided into three subsets as shown in table 3.1 and 3.2: 78 pairs (50%) for training, 48 pairs (30%) for testing and 24 pairs (20%) for validation. The training and testing subsets were used during the iterative weight optimisation process, while the validation subset, which was kept entirely separate and not used in any part of the training or testing procedure, was used as the final benchmark of model generalisation performance [26, 27].

Special attention was also given during dataset selection so that all deformation conditions were properly represented in the training, testing and validation groups. This helped in improving the stability, reliability and prediction accuracy of the developed ANN models for both alloy systems. The selected datasets included both lower and higher flow stress regions corresponding to different thermo-mechanical deformation conditions, which further improved the learning capability of the ANN network.

Table 3.1: Selection of data for Training, Testing and Validation of alloy A.

strain(mm/mm)	strain rate (per sec)	573	623	673	723	773
	0.001	193.3	171.5	140	108.5	77.5
	0.01	206.6	183.6	159.7	133.4	103.9
0.1	0.1	208.4	179.2	174.7	150.9	124.4
	1	203	195.7	187.2	175.6	145.6
	10	196.2	187.9	190.1	176.2	158.2
	0.001	212.1	183.1	144.3	107.7	76.7
	0.01	229.6	200.5	168.1	135.5	103.1
0.2	0.1	232.6	201.9	183.6	154.6	127.4
	1	227.6	214.3	200.3	187.4	147.5
	10	219.7	207	206.8	187.3	164.6
	0.001	217.5	185.6	140.4	104.5	73.3
	0.01	240.8	207.5	171.3	134.6	102
0.3	0.1	242.7	208.6	185.1	153.7	126.8
	1	236.6	219.3	201.8	188.3	141.9
	10	227.7	212.4	210.3	187.2	160.7
	0.001	216.5	184.7	136.5	99.9	69.2
	0.01	247.8	210.8	172.1	133.8	100.6
0.4	0.1	248.3	211.8	185.1	152.2	125.1
	1	242.1	220.9	200.7	186.7	137.1
	10	232	212.8	210.1	184.6	155.7
	0.001	214.5	178.9	129.4	92.7	63.6
	0.01	251.4	210.7	169.6	131.7	96.7
0.5	0.1	252.2	213.7	185.3	150.8	123.5
	1	246	221.7	199.4	183.7	133.7
	10	234.1	213.4	209.5	181.9	151.8
	0.001	212.7	175.8	124.7	88.4	59.7
	0.01	254.5	211.1	169.4	129.6	95.5
0.6	0.1	254.8	214.8	185.3	149.7	122.7
	1	248.4	222.2	198.5	180.4	131.5
	10	236.5	214.3	209.2	179.8	149.1

	Training		Testing		validation
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Table 3.2: Selection of data for Training, Testing and Validation of alloy B.

strain (mm/mm)	strain rate (per sec)	573	623	673	723	773
	0.001	191.1	161.9	133	97.3	75.4
	0.01	189.5	177.7	152.2	122.4	97.8
0.1	0.1	203.7	188	173	143.7	117.6
	1	194.7	197.3	182.4	161	135.9
	10	199.1	199.9	190	175.5	151.8
	0.001	205.9	167.6	133.5	96.3	73.3
	0.01	204.4	190.1	159.2	123	95.7
0.2	0.1	228.4	205.9	183.1	149.5	117.4
	1	217.5	215	191.5	164.5	136.8
	10	220.5	215.7	203	182	159.2
	0.001	209.6	165.6	130.3	93	70
	0.01	209.7	192.4	159.9	121.9	93.5
0.3	0.1	236.1	208.6	183.1	148.2	114.7
	1	226.3	217.7	189.6	159.2	131.3
	10	226.2	216.7	202.7	178.5	158.1
	0.001	210	162.4	126.4	90.2	66.7
	0.01	212.2	193.8	160.5	121.4	92.4
0.4	0.1	240.7	208.9	181	145.7	112
	1	228.7	217.4	185.6	154.3	126.5
	10	229	215.6	199.3	172.5	154.9
	0.001	207.3	156.6	118.6	85.3	61.7
	0.01	212.5	191.4	157.2	117	87.9
0.5	0.1	244.1	209.5	180.3	143.9	110.1
	1	230	216.5	182.3	150.5	123.3
	10	230.9	216	196	168.9	151
	0.001	205.7	152.2	113.3	81.3	58.5
	0.01	214.2	191.8	156.9	115.6	87.3
0.6	0.1	245.9	210.2	179.2	142.4	108.8
	1	230.4	215.6	180.1	148.3	120.9
	10	232.3	217	194.4	167.1	146.8

	Training		Testing		validation
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3.2 ANN Trial – Training and Testing

In the present work, several Artificial Neural Network (ANN) models were developed and tested separately for Alloy A and Alloy B in order to find the most suitable ANN architecture for accurate flow stress prediction. Since the deformation behaviour of aluminium alloys changes continuously with strain, strain rate and deformation temperature, a single ANN model was not sufficient for obtaining reliable prediction results. Therefore, many ANN trial models were generated and compared using a trial-and-error approach.

Different hidden neuron combinations and transfer function combinations were tested during the present investigation. Transfer functions such as tansig–purelin and logsig–purelin were mainly used for studying their effect on network prediction capability and convergence behaviour. All ANN models were developed using MATLAB Neural Network Toolbox with the Levenberg–Marquardt backpropagation training algorithm because of its faster convergence and stable training behaviour for non-linear engineering problems.

Before ANN training, the complete experimental dataset was divided into separate groups for training, testing and validation purposes. During training, the ANN models continuously adjusted their internal weights and biases in order to reduce the prediction error between experimental and predicted flow stress values. After completion of training, the developed ANN models were tested using unseen datasets to examine the prediction capability and generalization behaviour of the network under unknown deformation conditions.

It was observed that some ANN models produced lower training error but comparatively poor testing performance because of overfitting behaviour. Similarly, some ANN architectures showed unstable validation results even though their training performance appeared satisfactory. Therefore, proper comparison of maximum training error, maximum testing error, RMS validation value and maximum validation error was necessary for selecting the optimized ANN structure.

For Alloy A, several ANN trial model architectures were systematically tested under different deformation conditions, and the results of Alloy A are presented in Table no (3.3).

Table 3.3: Maximum absolute error, Maximum percentage error and RMS error obtained during training, testing and validation for Alloy A.

	Max. Error (MPa)	Max. % Error (MPa)	RMS Error (MPa)
Training	6.6179	7.1015	3.9866
Testing	13.1334	8.4106	6.3198
Validation	7.0143	4.4887	4.1287

Similarly, for Alloy B, different ANN model datasets shown in Table 3.2 were examined and compared.

Table 3.4: Maximum absolute error, Maximum percentage error and RMS error obtained during training, testing and validation for Alloy B.

	Max. Error (MPa)	Max. % Error (MPa)	RMS Error (MPa)
Training	9.0324	5.1671	3.4165
Testing	25.4754	11.9884	9.361
Validation	7.4115	3.9225	3.3381

From the ANN trial study, it was clearly observed that the prediction behaviour of the network strongly depends on hidden neuron selection and transfer function combination. The optimized ANN models selected for Alloy A and Alloy B shown in table no. 3.3 and 3.4 comparatively lower validation errors and better agreement between experimental and predicted flow stress values. These optimized ANN models were therefore further utilized for generation of large deformation datasets, contour map development and Deformation Mechanism Map (DMM) analysis.

3.3 Validation Results

The validation dataset for Alloy A consisted of 24 data points not used during training or testing. The optimised model predicted all 24 validation points within the $\pm 5\%$ deviation band as shown in fig. no (3.1) for both alloy A and alloy B, confirming satisfactory generalisation behaviour. The scatter plot of predicted versus experimental flow stress values showed that the data points cluster

closely around the line of perfect prediction, with no systematic bias across the range of flow stress values (approximately 50 to 250 MPa).

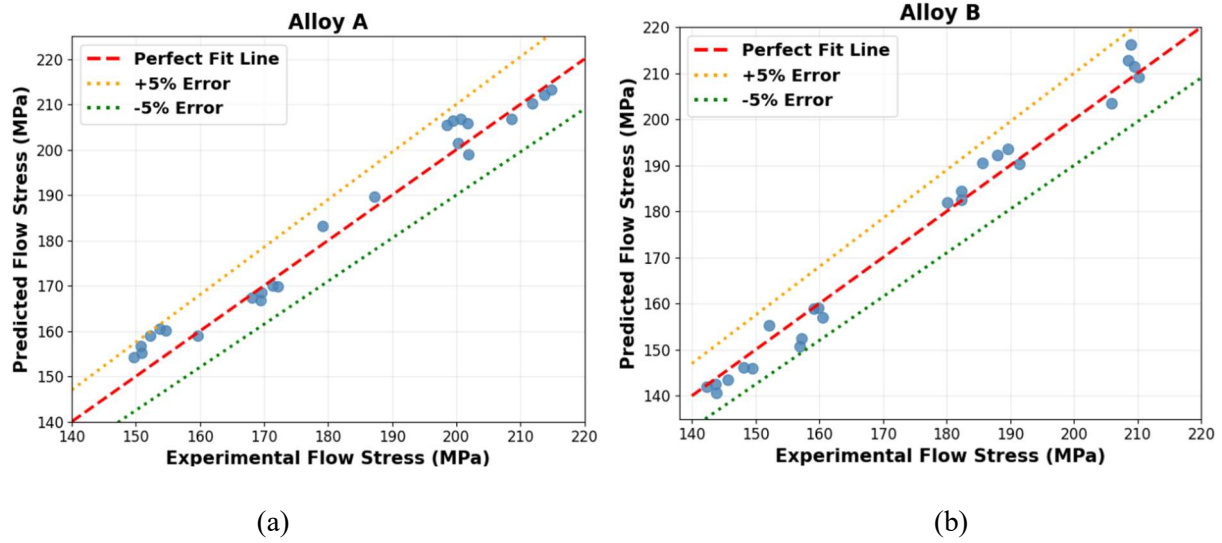


Figure 3.1: Comparison of predicted and experimental flow stress values for validation data

(a) Alloy A and (b) Alloy B.

Some architectures with lower training RMSD showed higher validation errors, confirming the overfitting concern mentioned earlier. The validation results made one thing quite clear — picking a model based purely on its training error is simply not good enough. A model genuinely earns its place only when it holds up consistently across all three subsets, and until it does that, it really cannot be trusted to generate the extended datasets that DMM analysis depends on [30].

An identical modelling procedure was followed for Alloy B, yielding the performance summary. The best-performing model for Alloy B, as shown in fig. no (3.1), used seven hidden neurons with logsig–purelin transfer functions and produced an RMS validation value of 0.0191 and a maximum validation error of 3.92%, which was lower than the corresponding figures for the Alloy A model.

The comparatively better performance of the Alloy B model may be attributed to the silver addition, which appears to reduce the variation in flow stress over the deformation range and thereby simplifies the learning process for the ANN model. It was also observed that the logsig transfer function consistently produced better validation performance than the tansig transfer function for most of the ANN architectures investigated in the present study [31,32].

3.4 Generation of Efficiency Power Dissipation

The efficiency contour maps developed in the present work helped in better understanding the hot deformation behaviour of Alloy A and Alloy B under different thermo-mechanical processing conditions [31]. These contour plots were generated using the ANN-predicted flow stress datasets over a wide range of deformation temperatures and strain-rate conditions. The maps provided useful insight into deformation stability, power dissipation behaviour and suitable hot working regions during deformation processing.

From the contour map analysis that is shown in fig. no. (3.2), it was clearly observed that the efficiency of power dissipation changes noticeably with deformation temperature and strain rate. In general, higher efficiency values were observed at elevated temperatures and lower strain-rate conditions. This behaviour indicates that the material undergoes comparatively smoother and more stable deformation at higher temperatures where recovery and thermal softening mechanisms become more effective [14, 15].

For Alloy A, the contour map showed relatively lower efficiency values at lower temperature and higher strain-rate regions, which indicates comparatively unstable deformation behaviour under those conditions. However, with increase in deformation temperature, the efficiency values gradually increased and larger stable deformation regions started appearing in the contour plots. The higher efficiency domains observed at elevated temperatures and lower strain rates may be related to improved hot workability and dynamic softening behaviour during deformation [14].

The contour map obtained for Alloy B also showed a similar overall trend, but comparatively improved efficiency distribution was observed over a wider deformation range. The addition of 0.1 wt.% silver appeared to improve the deformation stability of the alloy during hot working conditions. Higher efficiency regions were observed mainly at intermediate and higher temperature conditions, indicating comparatively better hot workability behaviour than the base alloy system.

It was also noticed that lower efficiency regions were mainly concentrated at higher strain-rate conditions for both alloy systems. Under such deformation conditions, the material deforms very rapidly and does not get sufficient time for recovery and microstructural stabilization processes to

occur properly. Because of this, unstable deformation behaviour becomes more dominant during hot working operations [14, 15].

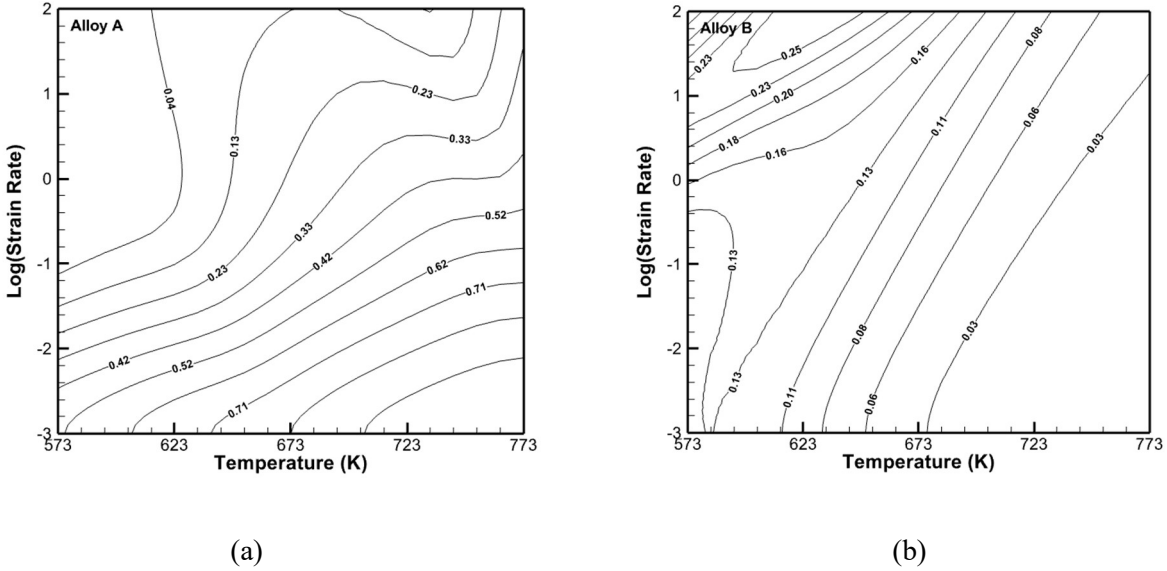


Figure 3.2: Efficiency power dissipation at strain 0.2 for (a) Alloy A and (b) Alloy B.

Overall, the generated efficiency contour maps successfully helped in identifying both stable and unstable deformation regions for Alloy A and Alloy B. The contour map analysis clearly demonstrated that ANN-generated datasets can effectively be utilized for hot workability analysis, deformation mechanism studies and selection of suitable thermo-mechanical processing conditions for aluminium alloy systems [14, 15, 31].

3.5 Instability Contour Map Analysis

The instability contour maps generated in the present work helped in analysing the unstable deformation behaviour of Alloy A and Alloy B under different thermo-mechanical processing conditions. The generated instability maps provided useful information regarding unsafe deformation regions where flow instability, localized deformation and non-uniform material flow may occur during hot working operations [35].

From the instability contour plots shown in fig. no. (3.3), it was observed that instability behaviour changes considerably with both deformation temperature and strain rate. In general, instability regions became more dominant at lower temperatures and higher strain-rate conditions [37].

For Alloy A, the instability contour map showed comparatively larger unstable regions at lower temperature and higher strain-rate conditions. These regions indicate comparatively poor deformation stability and higher chances of localized deformation behaviour during hot working. However, as the deformation temperature increased and strain rate decreased, the unstable regions reduced and comparatively safer deformation domains were observed in the contour map [35, 38].

The instability contour map generated for Alloy B showed comparatively smaller instability regions over several deformation conditions. The addition of 0.1 wt.% silver appeared to improve the deformation stability of the alloy system and reduced the tendency of unstable deformation behaviour during hot working [5]. More stable deformation regions were mainly observed at intermediate and higher temperature conditions with lower strain rates, indicating comparatively improved hot workability behaviour for Alloy B compared to the base alloy. It was also observed that instability regions became more concentrated at very high strain-rate conditions for both alloy systems [37].

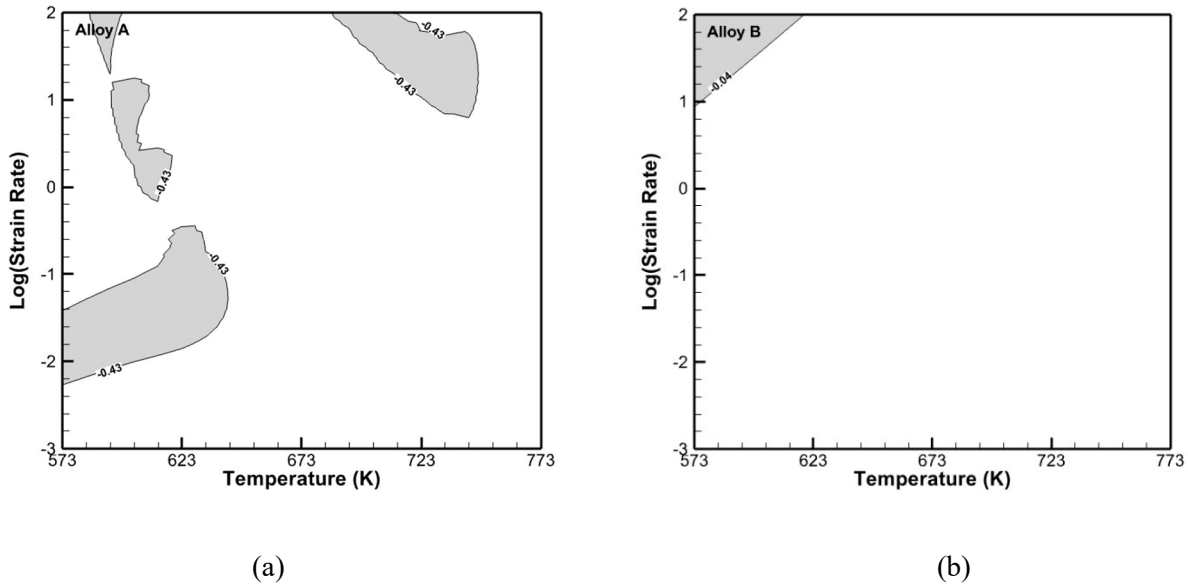


Figure 3.3: Instability plots at strain 0.2 for (a) Alloy A and (b) Alloy B.

Overall, the generated instability contour maps successfully helped in identifying unstable and safe deformation regions for both Alloy A and Alloy B. The contour map analysis clearly demonstrated that ANN-generated datasets can effectively be utilized for studying deformation instability

behaviour and for selecting suitable thermo-mechanical processing conditions during industrial hot working applications [35, 37, 38].

3.6 Generation of Deformation Mechanism Maps

After developing and validating the optimized ANN models, the predicted flow stress datasets were further used for generation of Deformation Mechanism Maps (DMM) for both Alloy A and Alloy B under different thermo-mechanical deformation conditions [33]. Since carrying out a very large number of hot deformation experiments experimentally is both difficult and time consuming, the ANN-generated datasets provided a simpler and more efficient approach for generating the large deformation database required for DMM analysis [31].

The generated datasets covered a wide range of strain (0.2, 0.4, 0.6), strain-rate and deformation temperature conditions for both alloy systems. Using these predicted flow stress values, contour maps corresponding to efficiency of power dissipation and instability behaviour were developed in order to study the deformation characteristics of the alloys during hot working conditions. These maps provided useful information regarding deformation stability, unsafe deformation regions and suitable processing domains during thermo-mechanical deformation [33].

From the generated Deformation Mechanism Maps (fig. no. 3.4, 3.5, 3.6), it was clearly observed that the deformation behaviour changes considerably with both deformation temperature and strain rate. In general, higher efficiency regions were mainly observed at elevated temperatures and lower strain-rate conditions, indicating comparatively stable deformation behaviour and better hot workability under such conditions. On the other hand, lower efficiency and larger instability regions were mostly concentrated at lower temperatures and higher strain-rate conditions where deformation occurs very rapidly with limited time available for recovery and microstructural stabilization processes [33, 34].

For Alloy A, comparatively stable deformation regions were mainly observed at higher deformation temperatures and lower strain-rate domains. However, at lower temperature and higher strain-rate conditions, larger instability regions appeared in the contour plots, indicating comparatively unstable deformation behaviour during hot working operations [33].

The generated DMM for Alloy B also followed a similar overall trend, but comparatively improved stability regions were observed under several deformation conditions. The addition of 0.1 wt.% silver appeared to improve the deformation stability and hot workability behaviour of the alloy

system. More stable deformation regions and comparatively reduced instability domains were observed for Alloy B under intermediate and higher temperature conditions.

The generated contour maps therefore helped in identifying suitable hot working conditions and unstable deformation regions for both alloy systems. Overall, the obtained results clearly demonstrated that ANN-generated datasets can effectively be utilized for Deformation Mechanism Map generation, hot workability analysis and selection of suitable thermo-mechanical processing conditions for aluminium alloy systems under elevated temperature deformation conditions [35].

Additionally, the generated Deformation Mechanism Maps as shown in fig. no. (3.4, 3.5, 3.6) that the hot deformation behaviour of both alloy systems changes noticeably with deformation temperature and strain-rate conditions. The contour maps provided a clearer understanding of the stable and unstable deformation regions, which becomes very helpful while selecting suitable hot working parameters during industrial processing operations. Therefore it demonstrate that ANN-assisted DMM analysis can be effectively used for understanding deformation behaviour and improving the hot workability characteristics of aluminium alloys [31, 33, 34].

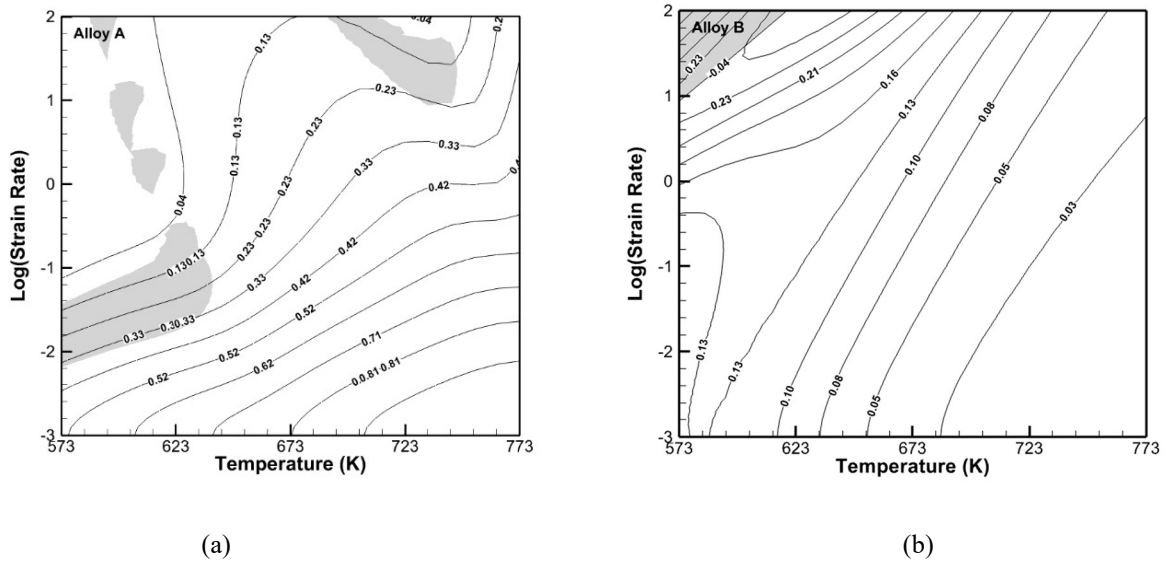


Figure 3.4: Deformation mechanism maps at a strain 0.2 for (a) Alloy A and (b) Alloy B.

620 K, instability regions are observed at both very low and very high strain-rate conditions. The instability region appearing near 580–610 K may be associated with localized flow and non-uniform deformation during hot working.

From Fig. 3.4 (b), corresponding to Alloy B at a strain of 0.2, stable deformation is mainly observed for temperatures above about 670 K and strain rates lower than $\log \dot{\epsilon} \approx 0.5$. The efficiency values vary between approximately 0.04 and 0.18. An instability region is observed at lower temperatures and higher strain-rate conditions, particularly below about 670 K. Compared to Alloy A, the instability domain is more concentrated and confined to a smaller processing window, indicating comparatively improved deformation stability due to silver addition.

In Fig. 3.5 (a), at a strain of 0.4, the stable deformation domain for Alloy A expands towards higher temperature regions. The efficiency values increase continuously and reach nearly 0.81 at temperatures above 720 K and lower strain-rate conditions. Small instability pockets are observed around 600–640 K under moderate strain-rate conditions. These regions indicate the possibility of unstable material flow during deformation.

In Fig. 3.5 (b), corresponding to Alloy B, the efficiency values remain in the range of approximately 0.04–0.18. Stable deformation is observed over a wider temperature range, whereas instability regions are mainly restricted to the low-temperature side of the map. The comparatively smaller instability domain suggests better hot workability of Alloy B under the investigated processing conditions.

From Fig. 3.6 (a), corresponding to Alloy A at a strain of 0.6, the stable deformation region is mainly observed for temperatures above about 650 K and strain rates lower than $\log \dot{\epsilon} \approx 0.5$. The efficiency of power dissipation reaches values as high as 0.95 in the high-temperature and low strain-rate domain. Some instability regions are observed near 620–670 K and at higher strain-rate conditions. These regions may indicate the onset of localized deformation during hot working.

From Fig. 3.6 (b), corresponding to Alloy B, stable deformation is observed over most of the investigated processing range. The instability region is mainly confined to the high-temperature and high strain-rate corner of the map. The efficiency values vary from approximately 0.02 to 0.18.

Compared to Alloy A, Alloy B exhibits a more uniform deformation domain and reduced instability tendency, indicating improved hot workability behaviour at higher strain levels.

Chapter 4: Conclusion

The present investigation developed ANN-based flow stress models for 2219 aluminium alloy Alloy A and Alloy B, and used those models to construct Deformation Mechanism Maps for both alloy systems. The principal conclusions drawn from the work are summarised as follows:

1. The ANN models was developed through trial and error method using feed-forward back propagation algorithm for both Alloy A and Alloy B.
2. The validation error remained within $\pm 5\%$, which shows that the developed ANN models were able to predict the flow stress values reasonably well.
3. Efficiency and instability contour maps were successfully generated using the ANN-predicted flow stress datasets, which helped in identifying stable and unstable deformation regions under different temperature and strain-rate conditions during hot working.
4. The Deformation Mechanism Maps (DMM) showed that stable deformation was generally observed at higher temperatures and lower strain-rate conditions, which helped in identifying suitable hot working regions for both alloy systems.
5. It was observed that the flow stress behaviour changes significantly with strain, strain rate and deformation temperature. Flow stress generally increased with increase in strain rate, whereas increasing deformation temperature reduced the flow stress because of thermal softening during hot deformation.

Chapter 5: Future Scope

The present work has established a working framework for ANN-based flow stress prediction and processing map development for 2219-series aluminium alloys. Several directions for future investigation follow naturally from the results and limitations of this study.

1. The prediction capability of the ANN model can be further improved by using a larger experimental dataset covering a wider range of temperatures and strain-rate conditions.
2. Other machine learning techniques such as Support Vector Regression (SVR), Random Forest, ANFIS and deep neural networks can be explored and compared with the present ANN model.
3. Future studies may include additional input parameters such as grain size, ageing condition and precipitate characteristics to improve flow stress prediction accuracy.
4. Detailed microstructural characterization can be carried out to validate the deformation mechanisms and hot workability domains identified from the DMM analysis.
5. The ANN–DMM methodology developed in this work can be extended to other engineering alloys, including magnesium, titanium and advanced aluminium alloy systems.
6. The developed ANN model can be integrated with finite element simulations of hot working processes for better process optimization and industrial applications.

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